



NCIE-007

# AI, Agent & Model Orchestration Specification

**In development — for human review — Version 1.1**

Specifies the orchestration layer beneath ARGUS: model routing, agents, Dynamic Agent Synthesis, tools, VPF output validation, delegation and failure containment

Model-independent: ARGUS is not a model, agent output is not institutional authority, and VPF validation is not human approval

Traces to NCIE-002 v0.4 (architecture), NCIE-003 v0.3 (canonical data), NCIE-004 v1.1 (engineering), NCIE-005 v1.2 (ARGUS behavior) and NCIE-006 v1.1 (Memory & Context)

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Maintained by: National Communications Intelligence Ecosystem (NCIE)

# Document Governance

|                            |                                                                                                                                                                                                |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document                   | NCIE-007 AI, Agent & Model Orchestration Specification                                                                                                                                         |
| Edition                    | In development — Version 1.1                                                                                                                                                                   |
| Baseline authorized        | NCIE-007 Skeletal v0.2 (VPF & Dynamic Agent Update) + NCIE-007-S01 v0.1 (Dynamic Agent Synthesis & Runtime Registration Supplement) — approved for full expansion; 36-chapter structure locked |
| Requirements authority     | NCIE-001 Master Product Requirements Document, Version 1.2                                                                                                                                     |
| Architecture authority     | NCIE-002 System Architecture & Technical Design, Version 0.4                                                                                                                                   |
| Canonical data authority   | NCIE-003 Data Model & Data Dictionary, Version 0.3 (Canonical Data Architecture Baseline)                                                                                                      |
| Engineering baseline       | NCIE-004 Technology Stack & Engineering Blueprint, Version 1.1 (Approved Engineering Baseline)                                                                                                 |
| ARGUS behavioral authority | NCIE-005 ARGUS Intelligence Assistant Specification, Version 1.2                                                                                                                               |
| Memory & Context authority | NCIE-006 Memory & Context Architecture Specification, Version 1.1                                                                                                                              |
| Owner                      | National Communications Intelligence Ecosystem (NCIE) Architecture Function                                                                                                                    |
| Reviewer                   | David King Boison (PhD) — Academic Intelligence Center                                                                                                                                         |
| Status                     | IN DEVELOPMENT — FOR HUMAN REVIEW until explicitly approved                                                                                                                                    |

## 0.1 Authority Chain

NCIE-001 requirements → NCIE-002 architecture → NCIE-003 canonical data → NCIE-004 engineering baseline → NCIE-005 ARGUS behavioral authority → NCIE-006 Memory & Context architecture → NCIE-007 AI/Agent/Model Orchestration. NCIE-007 implements these authorities; it does not reinterpret them. Where VPF or Dynamic Agent Synthesis introduces an impact on NCIE-001-006, that impact is recorded in Chapter 36's Upstream Impact Register for a surgical amendment pass after NCIE-007 is approved — the upstream documents are not amended during this production cycle.

## 0.2 Token-Conservation Convention

Every chapter opens with a short Upstream block citing the specific upstream chapter and stating only the orchestration consequence — never a restatement of the upstream architecture. Human-Primary governance, ARGUS behavior, Memory architecture, canonical data semantics and the engineering stack are not re-explained; each is referenced once and applied.

## 0.3 Governing Invariants Register

These 20 invariants bind every chapter of this document and are stated once here rather than re-derived repeatedly:

- ARGUS ≠ MODEL.
- ARGUS ≠ SPECIALIZED AGENT.
- MODEL ≠ INSTITUTIONAL AUTHORITY.
- AGENT ≠ INSTITUTIONAL DECISION MAKER.
- DELEGATION ≠ AUTHORITY EXPANSION.
- TOOL ACCESS ≠ ACTION AUTHORITY.
- MODEL AVAILABILITY ≠ MODEL ELIGIBILITY.
- AGENT CONSENSUS ≠ INSTITUTIONAL TRUTH.
- MODEL/AGENT OUTPUT ≠ EVIDENCE, FINDING OR DECISION.
- RETRIEVED CONTENT ≠ TRUSTED INSTRUCTION.
- ARGUS/AGENTS ≠ INSTITUTIONAL MEMORY.
- REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION.
- AGENT CREATION ≠ AUTHORITY CREATION.
- AGENT SYNTHESIS ≠ PERMISSION EXPANSION.
- EPHEMERAL AGENT ≠ REGISTERED INSTITUTIONAL AGENT.
- SYNTHESIZED AGENT ≠ AGENT-FACTORY AUTHORITY.
- AGENT SYNTHESIS ≠ SELF-REGISTRATION.
- VPF VALIDATION ≠ HUMAN / INSTITUTIONAL APPROVAL.

- UNVALIDATED AI OUTPUT ≠ VALIDATED NCIE AI RESULT.
- SUBSTANTIVE AI OUTPUT ≠ VPF-EXEMPT BY DEFAULT.

## 0.4 VPF and Dynamic Agent Synthesis — Scope Note

The VPF Validation Framework and Dynamic Agent Synthesis are controlled additions approved for inclusion in this edition (NCIE-007 Skeletal v0.2). VPF validates AI-generated output; it does not approve regulatory Decisions, create Findings, authorize blocking or tracking, approve Institutional Knowledge, or replace human judgment (VPF VALIDATION ≠ HUMAN / INSTITUTIONAL APPROVAL). Dynamic Agent Synthesis lets ARGUS create bounded task-specific agents from approved primitives without Codex hand-coding each one; it never creates new institutional authority, infrastructure or unapproved capability.

## 0.5 Human Review Focus Classification

Every chapter's Human Review Focus item carries one of seven labels: RESOLVED BY UPSTREAM BASELINE; PROPOSED DESIGN DEFAULT; INSTITUTIONAL CONFIRMATION REQUIRED; SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED; LEGAL/POLICY CONFIRMATION REQUIRED; ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED; or SOURCE/WORKFLOW DISCOVERY REQUIRED — each additionally marked Blocking or Non-Blocking, and each stating precisely what it blocks rather than implying the whole document is blocked. Chapter 35 consolidates every open item, extracted directly from the chapters so the register cannot drift from its sources (mechanically verified per §35.4).

## 0.6 Status Convention

Skeletal-structure approval authorizes expansion; it does not approve any individual architectural interpretation or Proposed Design Default. Every chapter in this edition is stamped IN DEVELOPMENT — FOR HUMAN REVIEW and remains so until explicitly approved. A technology named from NCIE-004 retains its existing NCIE-004 approval status here — using it in an NCIE-007 example never promotes it. Document completeness is not document approval.

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# Mandate, Authority Chain & Specification Boundary

Establish NCIE-007 as the orchestration authority beneath ARGUS without redefining upstream architecture.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4, Ch.24 — Human-Primary authority, Model Gateway, Tool Gateway and AI safety boundaries are architecturally fixed; NCIE-007 specifies their implementation. | NCIE-005 Ch.5-30 — ARGUS behavioral limits are canonical; NCIE-007 specifies what ARGUS orchestrates through, not how it behaves. | NCIE-006 Ch.1-36 — Memory/Context services are canonical; NCIE-007 specifies how agents consume them, never redefines them.

## 1.1 Mandate

NCIE-007 specifies the orchestration layer beneath ARGUS: model routing, specialized agents, Dynamic Agent Synthesis, delegation, tools, the VPF Validation Framework, provenance and failure containment. It implements upstream authority; it does not redesign Human-Primary governance, ARGUS's behavioral doctrine, canonical data semantics or Memory/Context architecture.

## 1.2 Specification Boundary — What NCIE-007 Does Not Own

Table 1. NCIE-007 specification boundary.

| Concern                                                                                  | Owned By                                                                                              |
|------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Institutional governance/human authority structures                                      | NCIE-008; NCIE-007 specifies what orchestration must expose to it, not the authority structure itself |
| Identity, authorization enforcement engineering, sandbox/network security implementation | NCIE-009; NCIE-007 specifies authorization/isolation requirements, not IAM/security implementation    |
| Evidence object lifecycle, chain-of-custody engineering                                  | NCIE-003 Ch.17; NCIE-010; NCIE-007 references Evidence, never redefines it                            |
| API/event transport engineering                                                          | NCIE-011; NCIE-007 specifies gateway/service contracts, not wire protocols                            |
| Physical compute/storage/model-hosting technology selection                              | NCIE-004 (Technology Decision Matrix); NCIE-014; NCIE-007 specifies logical roles only                |
| User experience of agent/VPF status to end users                                         | NCIE-012; NCIE-007 specifies the states to be shown, not the UI                                       |
| Comprehensive test-suite implementation                                                  | NCIE-016; NCIE-007 Ch.32 specifies required test categories, not the full verification programme      |
| ARGUS conversational/analytical/voice behavior                                           | NCIE-005; NCIE-007 specifies what ARGUS may orchestrate, not how ARGUS behaves                        |
| Memory/Context persistence and Institutional Knowledge lifecycle                         | NCIE-006; NCIE-007 specifies how agents consume governed Memory/Context, never redefines it           |

## 1.3 Structural Change Control

The 36-chapter structure is locked. A material contradiction or omission discovered during expansion is flagged STRUCTURAL CHANGE REQUIRED — HUMAN REVIEW rather than silently resolved. None has been raised in this edition.

## Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the acceptance authority for NCIE-007 as a whole and for individual chapter sign-off.

Traceability: NCIE-002 Ch.4, Ch.24; NCIE-005; NCIE-006.

# Orchestration Doctrine & Governing Invariants

*Lock the invariants governing models, agents, delegation, tools, VPF and Dynamic Agent Synthesis.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — ARGUS ≠ institutional authority is architecturally fixed; this chapter locks the orchestration-specific corollaries. | NCIE-006 front matter §0.3 — REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION and PRIOR HUMAN DECISION ≠ CURRENT ACTION AUTHORITY apply identically to every model/agent invocation.

## 2.1 Authoritative Definitions

Table 2. Core term definitions, stated once and referenced throughout this document.

| Term                     | Definition                                                                                                                                                                                 |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Model                    | A hosted AI model/deployment invoked only through the Model Gateway (Ch.5); a replaceable implementation component with no institutional standing of its own.                              |
| Agent                    | A bounded runtime actor — Registered Reusable or Ephemeral Synthesized (Ch.11) — that performs a declared task using approved models/tools/context; never an institutional decision maker. |
| Tool                     | A governed capability invoked only through the Tool Gateway (Ch.19); read-only tools are potentially consequential (Ch.20 §20.2).                                                          |
| Delegation               | The act of a human, ARGUS or an agent assigning a bounded subtask to another actor; never a mechanism for expanding authority (Ch.14).                                                     |
| VPF Disposition          | The explicit validation outcome (PASS/QUALIFIED/REJECT/HUMAN_REVIEW_REQUIRED, Ch.26) an AI-generated output receives before validated release; a disposition, not an approval.             |
| Agent Definition Package | The canonical logical structure describing a synthesized agent's purpose, scope, permitted primitives and constraints (Ch.11 §11.4).                                                       |

## 2.2 Governing Invariants

The 19 invariants in front matter §0.3 bind every chapter without exception, including the controlled additions for VPF and Dynamic Agent Synthesis. None is re-derived per chapter; each chapter applies the subset relevant to its scope.

## 2.3 Model Independence

Replacing the ARGUS model/provider, any routed model, an agent runtime or an agent framework never transfers institutional authority to that replacement and never resets Institutional Memory (NCIE-006 §2.3). Orchestration doctrine belongs to NCIE, not to any model vendor's proprietary agent, tool-calling or validation feature (Ch.36 sovereignty acceptance tests).

## Human Review Focus

- [RESOLVED BY UPSTREAM BASELINE]** All 19 governing invariants are fixed by NCIE-002 Ch.4 and the approved NCIE-007 Skeletal v0.2; no institutional decision is required to adopt them.

Traceability: NCIE-002 Ch.4; NCIE-006 front matter §0.3.

# AI Runtime Actor & Capability Taxonomy

Define runtime actor types (Human, ARGUS, Registered Agent, Ephemeral Agent, Model, Tool, Service) and machine-readable capability classes.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

Upstream: NCIE-002 Ch.4 — Zero-trust, distinct-identity requirements apply to every orchestration actor without exception.

## 3.1 Runtime Actor Taxonomy

Figure 1. AI Runtime Actor Taxonomy — every instantiated actor requires a distinct runtime identity (§0.3; audit must identify which actor performed each operation, Ch.27).

| SOURCE            | RELATIONSHIP                          | TARGET                                      |
|-------------------|---------------------------------------|---------------------------------------------|
| Human Principal   | is never assumed to be                | ARGUS                                       |
| ARGUS             | coordinates but is never identical to | Specialized Agent (Registered or Ephemeral) |
| Specialized Agent | invokes but is never identical to     | Model                                       |
| Model / Agent     | invokes but is never identical to     | Tool                                        |
| Any Runtime Actor | is never identical to                 | Workflow or Service Identity                |

## 3.2 Capability Primitive Classes

Every orchestration capability is composed only from approved or eligible primitives:

- Models — eligible hosted deployments invoked via the Model Gateway (Ch.5-6).
- Tools — governed capabilities invoked via the Tool Gateway (Ch.19).
- Context Services — governed NCIE-006 Memory/Context interfaces (Ch.17).
- Schemas — governed output/input contracts (Ch.24, Ch.26).
- Policies — governed routing, risk, VPF and resource policy (Ch.7, Ch.20, Ch.26, Ch.29).
- Execution Runtimes — sandboxed environments bounding agent execution (Ch.34).

ARGUS and the Agent Factory (Ch.12) compose capabilities only from these approved primitives; neither synthesizes new permissions or infrastructure merely because a task requires them.

Traceability: NCIE-002 Ch.4.

# ARGUS Orchestrator Boundary

Define how ARGUS coordinates AI capabilities without becoming the Model Gateway, Memory Service or Tool Gateway.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-005 Ch.4 — ARGUS is a Human-Primary conversational/analytical assistant, not a model, agent or institutional authority; this chapter specifies its orchestration boundary.

## 4.1 What ARGUS Coordinates, Not Owns

Table 3. ARGUS Orchestrator Boundary — ARGUS requests and coordinates; the governed services decide and execute.

| ARGUS Coordinates                                                 | ARGUS Does Not Become                                                                                       |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| Model selection request                                           | The Model Gateway (Ch.5) — routing/eligibility decisions are made there, not by ARGUS directly              |
| Agent task assignment, including Dynamic Agent Synthesis requests | The Agent Factory (Ch.12) — synthesis, sandboxing and Gate A occur there                                    |
| Tool invocation request                                           | The Tool Gateway (Ch.19) — authorization and execution occur there                                          |
| Context retrieval request                                         | The NCIE-006 Memory/Context Service — ARGUS never owns Institutional Memory directly                        |
| AI output presentation                                            | The VPF Validation Framework (Ch.26) — ARGUS never self-certifies its own or an agent's output as validated |

## 4.2 Orchestration Request Lifecycle

Figure 2. ARGUS Orchestration Request Lifecycle — every path terminates in a governed service, never an ARGUS-internal shortcut.

|    |                                                                                          |
|----|------------------------------------------------------------------------------------------|
| 01 | Human or governed workflow request                                                       |
| 02 | ARGUS interprets task/purpose                                                            |
| 03 | Existing-capability suitability check (registered agent, direct model, or governed tool) |
| 04 | Route to Model Gateway / Tool Gateway / Agent Factory as applicable                      |
| 05 | Result returned to ARGUS                                                                 |
| 06 | VPF Gate B validation where applicable (Ch.26)                                           |
| 07 | ARGUS presents result with disposition to the human                                      |

Traceability: NCIE-005 Ch.4.

# Model Gateway Contract

Define the single governed entry boundary between ARGUS/agents and eligible model deployments.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.24 — The Model Gateway is the single governed entry boundary to eligible model deployments; this chapter specifies its contract.

## 5.1 Model Gateway Mandate

All model access remains through the Model Gateway. Dynamic agents and registered agents alike shall not embed direct provider SDK dependencies as an escape path. This applies to every runtime actor defined in Ch.3, without exception for ephemeral or synthesized agents.

## 5.2 Routing Priority

Model selection is Eligibility First, then Capability / Performance / Cost / Latency optimization (Ch.7 defines eligibility criteria in full). An ineligible model is never used as fallback merely because an eligible model is temporarily unavailable (Ch.10 §10.2).

## 5.3 Gateway Contract

Table 4. Model Gateway request/response contract.

| Element            | Purpose                                                                                                |
|--------------------|--------------------------------------------------------------------------------------------------------|
| Request            | Task profile, purpose, classification, residency, requesting actor identity (Ch.3).                    |
| Eligibility Result | Pass/fail against Ch.7's criteria before routing proceeds.                                             |
| Route              | The selected eligible model/deployment, or a degraded/no-route state (Ch.10).                          |
| Response           | Model output plus route/version metadata required for provenance (Ch.9, Ch.27).                        |
| Telemetry          | Latency, token/cost and error taxonomy (Ch.31) — operational, never authoritative audit (Ch.31 §31.4). |

Traceability: NCIE-002 Ch.24.

# Approved Model Registry

Define governed metadata and lifecycle for models/deployments eligible for NCIE use.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-004 Ch.18 — Model/provider approval and lifecycle are engineering-governed; this chapter specifies the orchestration-facing registry contract.

## 6.1 Registry Fields

Table 5. Approved Model Registry fields — evaluated per Ch.7's eligibility dimensions before routing.

| Field                  | Purpose                                                                                            |
|------------------------|----------------------------------------------------------------------------------------------------|
| Model ID / Deployment  | Stable reference distinguishing model version from hosting deployment.                             |
| Provider               | The eligible provider hosting this deployment (Ch.8 governs adapter isolation).                    |
| Capabilities           | Declared task/tool/output capabilities of this deployment.                                         |
| Classification Ceiling | The maximum data classification this deployment may process.                                       |
| Residency              | Approved data-residency scope for this deployment.                                                 |
| Retention              | Provider-side retention behavior applicable to requests/outputs.                                   |
| Lifecycle State        | Candidate / Evaluated / Approved / Active / Restricted / Suspended / Deprecated / Retired (Ch.33). |

## 6.2 Registry Mutation Boundary

AI components cannot directly mutate Registry state outside authorized lifecycle workflows (Ch.33). Routing honors current Registry state immediately upon a lifecycle change — a suspended model is never used merely because a cached eligibility result has not expired.

### Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the model-approval authority and cadence for Registry review.
- Traceability: NCIE-004 Ch. 18.

# Classification-, Purpose- & Task-Aware Model Routing

Define deterministic eligibility and routing based on classification, purpose, task and policy.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-003 Ch.31 — Classification/residency/retention taxonomy is canonical; this chapter applies it to model eligibility.

## 7.1 Eligibility Dimensions

Table 6. Model routing eligibility dimensions, evaluated in order — availability is never allowed to promote an ineligible model.

| Dimension            | Evaluated Before Routing                                                                                 |
|----------------------|----------------------------------------------------------------------------------------------------------|
| Classification       | Model's classification ceiling (Ch.6) must meet or exceed the task's classification.                     |
| Purpose              | Model's approved purpose scope must cover the requested task type.                                       |
| Task                 | Model's declared capabilities must cover the specific task (e.g., structured output, tool use).          |
| Residency            | Model deployment's data-residency scope must satisfy the request's residency requirement.                |
| Provider Eligibility | Provider must hold current approved status (Ch.8).                                                       |
| Retention            | Provider-side retention behavior must satisfy the request's retention requirement.                       |
| Model Capability     | Declared model capability must be sufficient for the task profile.                                       |
| Tool Requirements    | Where the task requires tool-calling, the model must support the governed tool-calling contract (Ch.19). |
| Availability         | Evaluated last — availability alone never substitutes for the preceding eligibility checks.              |

## 7.2 Routing Result

A routing request yields an eligible route, or an explicit degraded/no-route state (Ch.10). MODEL AVAILABILITY ≠ MODEL ELIGIBILITY (front matter §0.3) — an available but ineligible model is never selected merely because it responds.

Traceability: NCIE-003 Ch.31.

## Provider Adapters & Provider Neutrality

*Isolate provider-specific APIs behind replaceable adapters.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-004 Ch.18 — Provider neutrality is an engineering-governed principle; this chapter specifies the orchestration-facing adapter boundary.

### 8.1 Adapter Isolation

Provider-specific APIs, request/response formats and proprietary tool-calling conventions are isolated behind replaceable adapters. No provider-specific capability becomes a hidden architectural dependency that undermines ARGUS continuity, Agent Definition portability, VPF sovereignty, Memory sovereignty or Tool governance (Ch.36 sovereignty acceptance tests).

### 8.2 Provider Outage Behavior

A provider outage never causes classification downgrade, residency bypass, retention-policy bypass or use of an unapproved provider. Where no eligible route remains, the Gateway returns a degraded/no-route state (Ch.10) rather than silently relaxing eligibility.

*Traceability: NCIE-004 Ch. 18.*

## Model Selection, Versioning & Reproducibility

*Preserve exact model/deployment context for consequential persisted outputs.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-003 Ch.28 — AI provenance/model-route semantics are canonical; this chapter specifies what must be preserved at selection time.

### 9.1 Persisted Selection Context

For consequential persisted output, the Model Gateway persists sufficient context to identify:

- Model Route and Deployment.
- Model Version/Identifier, where required by policy.
- Applicable Routing Policy Version.
- Invocation Time.

Private chain-of-thought is never persisted. A later model-version change is recorded as a new route on new outputs; historical outputs retain their original route reference (Ch.33 §33.4 model-change acceptance behavior).

*Traceability: NCIE-003 Ch.28.*

# Fallback, Failover & Degraded AI Operation

Define safe fallback and degraded operation without weakening eligibility controls.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.24 — AI safety and service-boundary requirements govern degraded operation; this chapter specifies fallback behavior.

## 10.1 Degraded-Mode Behavior

Table 7. Fallback/failover behavior — eligibility constraints are never relaxed to preserve availability.

| Condition                                        | Behavior                                                                                                                                |
|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| No eligible model for a request                  | Return degraded/no-route state; never substitute an ineligible model                                                                    |
| Preferred eligible model unavailable             | Fall back only to another currently eligible model; ineligible models are never used as fallback                                        |
| All routes for a classification tier unavailable | Deterministic NCIE functions continue independently — AI orchestration failure never disables non-AI regulatory functions (Ch.34 §34.3) |

## 10.2 Model Suspension Propagation

A model/deployment may become temporarily ineligible even if technically available (e.g., suspected compromise, Ch.33 §33.5). Routing honors current Registry state immediately; a suspended model is never used through a stale cached eligibility result.

Traceability: NCIE-002 Ch.24.

# Specialized Agent Architecture & Dynamic Agent Synthesis

Define bounded specialized agents — Registered Reusable and Ephemeral Synthesized — and the Dynamic Agent Synthesis capability that creates the latter.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007-S01 §3-5 — Agent Definition Package structure and the DRAFT→ACTIVE\_REUSABLE lifecycle are the normative supplement baseline this chapter implements in full. | NCIE-005 Ch.4 — ARGUS may initiate bounded synthesis; ARGUS never grants an agent new institutional authority.

## 11.1 Two Agent Kinds

Specialized agents are either Registered Reusable Agents (approved for repeated institutional use, Ch.12) or Ephemeral Synthesized Agents (instantiated from a validated Agent Definition Package for one bounded task, expiring by policy unless promoted). EPHEMERAL AGENT ≠ REGISTERED INSTITUTIONAL AGENT (front matter §0.3) — neither kind is an institutional decision maker.

## 11.2 Dynamic Agent Synthesis Flow

Figure 3. Dynamic Agent Synthesis Flow — this capability must be fully implementable from NCIE-007 without returning to Codex for bespoke development of each bounded agent.

|    |                                                                                                                                     |
|----|-------------------------------------------------------------------------------------------------------------------------------------|
| 01 | Task need identified — existing registered/eligible agent suitability check                                                         |
| 02 | No adequate existing agent — ARGUS requests synthesis; Agent Definition Package generated from approved primitives only (Ch.3 §3.2) |
| 03 | VPF Gate A — Agent Envelope Validation (§11.5)                                                                                      |
| 04 | Security/policy validation and sandbox evaluation (Ch.34 §34.4)                                                                     |
| 05 | Ephemeral Approval — agent activates only within its validated envelope and current authorization context                           |
| 06 | Execution                                                                                                                           |
| 07 | VPF Gate B — AI Output Validation (Ch.26 §26.4) before ARGUS presents a validated NCIE AI result                                    |
| 08 | Ephemeral agent expires per policy unless it becomes a Promotion Candidate (§11.7)                                                  |

## 11.3 Existing-Agent Reuse Before Synthesis

ARGUS prefers or at least identifies existing registered-agent capability reuse before synthesizing a duplicate. Where no approved primitive can satisfy the task, ARGUS produces an ENGINEERING CHANGE CANDIDATE rather than fabricating the missing capability (§11.6).

## 11.4 Agent Definition Package

The canonical logical structure for a synthesized Agent Definition Package, at minimum:

- Agent Definition ID; Purpose; Task Scope.
- Input Schema; Output Schema.
- Permitted Models; Permitted Tools; Permitted Data Classes.
- Context Scope; Authorization Envelope; Delegation Depth.
- Resource/Token/Time Budget; Network/Egress Policy; Persistence Policy.

- VPF Validation Profile; Evaluation Tests.
- Termination/Expiry Conditions; Provenance Requirements.

No physical database fields are invented beyond what is logically required; NCIE-014 specifies physical storage.

11.5 VPF Gate A — Agent Envelope Validation

Before an agent executes, Gate A validates its Purpose, Task Scope, Inputs/Outputs, Models, Tools, Data Classes, Context Scope, Authorization Envelope, Delegation Depth, Network/Egress, Persistence, Resource Budget, Termination Conditions and Policy Boundaries. Gate A and Gate B (Ch.26 §26.4) are never collapsed into one check.

11.6 Approved Primitives Only

Dynamic agents are composed only from approved/eligible Models, Tools, Context Services, Schemas, Policies, Execution Runtimes and Capability Primitives (Ch.3 §3.2). ARGUS shall not synthesize new permissions or infrastructure merely because a task requires them. A task requiring a capability outside approved primitives yields an ENGINEERING CHANGE CANDIDATE, not a fabricated one.

11.7 Ephemeral Agent Lifecycle

Figure 4. Ephemeral Agent Lifecycle — state names may be normalized in implementation but semantics remain explicit.

|    |                                                                                            |
|----|--------------------------------------------------------------------------------------------|
| 01 | DRAFT                                                                                      |
| 02 | SYNTHESIZED                                                                                |
| 03 | SANDBOXED                                                                                  |
| 04 | VPF_VALIDATED                                                                              |
| 05 | EVALUATED                                                                                  |
| 06 | EPHEMERAL_APPROVED                                                                         |
| 07 | ACTIVE_EPHEMERAL                                                                           |
| 08 | EXPIRED — with exceptional states REJECTED, SUSPENDED or TERMINATED available at any stage |

11.8 Reusable-Agent Promotion

A repeatedly useful ephemeral agent may become a PROMOTION\_CANDIDATE → HUMAN/GOVERNANCE REVIEW → REGISTERED → ACTIVE\_REUSABLE. ARGUS may recommend promotion; ARGUS shall not approve its own agent for permanent institutional registration (AGENT SYNTHESIS ≠ SELF-REGISTRATION, front matter §0.3).

11.9 Current Authorization Envelope

Every synthesized agent executes under a current authorization envelope derived from the active governed principal/task context. It shall not obtain authority from Historical Memory, Prior Human Decision, ARGUS Recommendation, Agent Definition Text or another agent's claimed permission (REMEMBERED AUTHORIZATION ≠ CURRENT AUTHORIZATION; PRIOR HUMAN DECISION ≠ CURRENT ACTION AUTHORITY).

Human Review Focus

- [INSTITUTIONAL CONFIRMATION REQUIRED] Approve the ephemeral-agent activation policy by task/risk class — whether ephemeral agents may activate automatically after Gate A plus security/evaluation checks, or require human approval per risk class.
- [INSTITUTIONAL CONFIRMATION REQUIRED] Approve the promotion authority and criteria for converting an ephemeral agent into a registered reusable institutional agent.

Traceability: NCIE-007-S01 §3-5; NCIE-005 Ch.4.

# Agent Capability Registry & Agent Factory

Define the governed registry of agent capabilities and the Agent Factory that synthesizes bounded agents from approved primitives.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007-S01 §1, §7 — Codex builds and maintains the Agent Factory; ARGUS uses it. Promotion requires governed review and explicit registration authority.

## 12.1 Agent Factory Boundary

CODEx BUILDS THE AGENT FACTORY; ARGUS USES THE AGENT FACTORY. Codex does not need to hand-code every future bounded task-specific agent once the Agent Factory exists. ARGUS does not become a general unrestricted software-development authority. Codex/engineering re-enters when a required capability exceeds the approved Agent Factory primitives or requires material production engineering change — a new Tool, Connector, Infrastructure, Security Primitive, Persistent Service or permanent generated-code integration.

## 12.2 Registry Distinctions

Table 8. Registry distinctions — shared storage implementation never collapses their governance semantics.

| Registry               | Governs                                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Agent Registry         | Reusable Agent Definitions, lifecycle state, promotion history — distinct from ephemeral run-scoped definitions.                                                              |
| Model Registry         | Eligible models/deployments (Ch.6) — distinct governance from Agent Registry.                                                                                                 |
| Tool Registry          | Eligible tools (Ch.19) — distinct governance from Agent Registry.                                                                                                             |
| VPF Profile Registry   | Eligible VPF validation profiles (Ch.26 §26.3) — distinct governance from Agent Registry.                                                                                     |
| Agent Pattern Registry | Approved reusable synthesis patterns — distinct governance, and existing historical agent instances retain their original pattern reference when a pattern is later modified. |

## 12.3 Recursive Synthesis Boundary

A synthesized agent shall not automatically receive Agent Factory access. SYNTHESIZED AGENT ≠ AGENT-FACTORY AUTHORITY (front matter §0.3). Default delegation depth for agent creation is ARGUS → Synthesized Agent, never ARGUS → Agent A → Agent B → Agent C. Unless Human Review explicitly changes it: RECURSIVE AGENT SYNTHESIS REMAINS DISABLED.

## 12.4 Registry Mutation Boundary

AI components — ARGUS, agents, the Agent Factory itself, VPF — cannot directly mutate governed Registry state outside authorized lifecycle workflows. Promotion to reusable registration considers Need, Evaluation, VPF History, Security, Ownership, Maintainability, Scope and Human/Governance Approval; repeated use or repeated VPF PASS alone never triggers automatic promotion.

## Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED] Approve the Agent Factory resource, model, tool, data and network ceilings applied to every synthesized agent.
- [PROPOSED DESIGN DEFAULT] Recursive Agent Synthesis remains disabled by default pending any future explicit governed policy change, consistent with NCIE-007-S01 §8.

Traceability: NCIE-007-S01 §1, §7-8.

# Task Decomposition & Delegation

Define decomposition of complex work into bounded delegated subtasks, including subtasks satisfied by Dynamic Agent Synthesis.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — Delegation ≠ authority expansion is architecturally fixed; this chapter specifies bounded subtask decomposition.

## 13.1 Decomposition Boundary

A complex task decomposes into bounded delegated subtasks. Where no eligible existing agent satisfies a bounded subtask, ARGUS may request Dynamic Agent Synthesis (Ch.11). The synthesized agent receives only the task-specific subset of context, tools, data classes and authority required for that subtask — never the parent task's full scope by default.

## 13.2 Subtask Record

Table 9. Subtask decomposition record.

| Field            | Purpose                                                             |
|------------------|---------------------------------------------------------------------|
| Parent Task      | The originating task this subtask decomposes from.                  |
| Subtask          | The bounded unit of work assigned.                                  |
| Delegator        | The human, ARGUS or agent that issued the delegation.               |
| Purpose          | Why this subtask exists (feeds minimum-necessary context, Ch.17).   |
| Context Subset   | The bounded context the subtask actually receives.                  |
| Constraints      | Resource, tool, data-class and temporal constraints (Ch.14, Ch.29). |
| Completion State | Pending / Completed / Failed / Escalated.                           |

Traceability: NCIE-002 Ch.4.

# Delegated Authorization & Identity Propagation

*Propagate current authorization without allowing delegation or synthesis to expand authority.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-006 Ch.25 §25.3 — Remembered Authorization ≠ Current Authorization applies identically to every delegated/synthesized agent invocation.

## 14.1 Current Authorization Propagation

Delegated and synthesized agents propagate current authorization; they never expand it. A synthesized agent has its own runtime identity (Ch.3 §3.1) and effective authorization envelope — it cannot inherit broader ARGUS privileges, another user's private context, or create new permissions through its generated definition.

## 14.2 Authorization Recheck

Table 10. Delegated authorization propagation model.

| Element                       | Behavior                                                                                                        |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Human Principal               | The accountable current human whose authorization anchors the delegation.                                       |
| Agent/Service Identity        | The distinct runtime identity executing the delegated work (never conflated with the Human Principal or ARGUS). |
| Current Authorization Context | Rechecked at time of use, not assumed from delegation time.                                                     |
| Purpose                       | The declared reason authorization was granted; expiring when the purpose completes.                             |
| Expiry / Recheck / Denial     | A delegated authorization that has expired or been revoked is denied, never silently honored from cached state. |

## 14.3 Recursive Delegation Boundary

Delegation cannot create transitive privilege expansion. If agent-to-agent delegation is permitted (Ch.22), every hop preserves Parent Task, Purpose, Effective Principal, Classification, Context Subset, Tool Ceiling, Resource Ceiling and Provenance — no hop widens them.

### Human Review Focus

- [SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED]** Confirm the authorization-revocation propagation latency acceptable for an active agent run — historical authorization shall not remain effective through cached agent state.

Traceability: NCIE-006 Ch.25 §25.3.

# Multi-Agent Coordination & Conflict Handling

*Coordinate multiple agents — registered and ephemeral — while preserving conflicts, attribution and human control.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — Agent consensus ≠ institutional truth is architecturally fixed; this chapter specifies multi-agent disagreement handling.

## 15.1 Disagreement Is Not Resolved by Voting

Multi-agent disagreement — among registered agents, ephemeral synthesized agents, or a mix — is never resolved through simple majority voting. AGENT CONSENSUS ≠ INSTITUTIONAL TRUTH (front matter §0.3).

## 15.2 Preserved Disagreement Record

- Agent A Result; Agent B Result (and further agents where applicable).
- Evidence Basis for each result.
- Contradictions between results.
- Uncertainty in each result.
- VPF Disposition for each result (Ch.26).

ARGUS may synthesize the disagreement for humans — presenting the record above — but never collapses it into a single asserted truth on the agents' behalf.

*Traceability: NCIE-002 Ch.4.*

# Agent State, Checkpointing & Resumption

*Persist operational run/checkpoint state without turning agent state into Institutional Memory.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-006 Ch.20 — Agent state is not Institutional Memory; material reasoning artifacts requiring institutional persistence flow through governed NCIE-006 services.

## 16.1 Run-Scoped State

Synthesized-agent state is run-scoped by default. Checkpoints preserve the Agent Definition version and VPF Gate A result in force at checkpoint time. Expired ephemeral agents shall not silently reactivate as reusable capabilities — reactivation always re-evaluates current authorization and eligibility (Ch.11 §11.9).

## 16.2 Memory Independence

Agent checkpoints may preserve operational run state; they are not Institutional Memory. Material reasoning artifacts that require institutional persistence — Findings, Decisions, Knowledge Candidates — are stored through the governed NCIE-006 Memory/Context architecture, never left only in agent checkpoint storage. Agent expiry shall not erase properly persisted institutional artifacts.

## 16.3 Checkpoint Fields

Table 11. Agent run/checkpoint record.

| Field             | Purpose                                                                                     |
|-------------------|---------------------------------------------------------------------------------------------|
| Run ID            | Stable reference for this specific agent execution.                                         |
| Checkpoint        | Point-in-time operational state snapshot.                                                   |
| Task State        | Progress against the declared task/subtask.                                                 |
| Context Reference | Pointer to the governed context the run consumed, not a private copy.                       |
| Retry / Resume    | Whether this run may be retried or resumed and under what current-authorization conditions. |
| Expiry            | The governed expiry condition applicable to this run.                                       |
| Terminal State    | Completed / Failed / Expired / Terminated.                                                  |

## Human Review Focus

- [ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED]** Confirm checkpoint retention duration and storage tier separate from institutional Memory retention (NCIE-006 Ch.28).

Traceability: NCIE-006 Ch.20.

# Context Acquisition & Memory-Service Integration

Acquire minimum-necessary governed context through NCIE-006 Memory/Context services.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-006 Ch.14, Ch.17 — Minimum-necessary retrieval and the governed ARGUS Memory Retrieval Interface are canonical; this chapter is their orchestration-side consumption contract.

## 17.1 Minimum-Necessary Context Acquisition

Synthesized and registered agents receive only the context required for their declared task. They consume governed context exclusively through NCIE-006 services and shall not inherit ARGUS's entire context, another user's private context, unrestricted Institutional Memory, or all Room information merely because ARGUS created or coordinates them.

## 17.2 Context Request Contract

Table 12. Context acquisition contract — mirrors NCIE-006's Context Envelope exactly, never a parallel structure.

| Field                | Purpose                                                                                     |
|----------------------|---------------------------------------------------------------------------------------------|
| Context Request      | Declares the requesting actor identity (Ch.3), purpose and task scope.                      |
| Context Envelope     | The NCIE-006 Context Envelope (NCIE-006 Ch.4) returned, bounded to the declared purpose.    |
| Temporal Mode        | Current / Historical / Decision-Time — never left implicit (NCIE-006 Ch.12-13).             |
| Classification       | The governing classification of the returned context.                                       |
| Authorization        | The effective authorization the context was assembled under.                                |
| Reconciliation State | Whether the returned context is Current, Stale or requires Reconciliation (NCIE-006 Ch.13). |

Traceability: NCIE-006 Ch. 14, Ch. 17.

# Evidence & Grounding Integration

*Ground model/agent work in authorized Evidence while preserving epistemic boundaries.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-003 Ch.17 — Evidence lifecycle and chain-of-custody are canonical; this chapter governs how models/agents ground work in it.

## 18.1 Grounding Boundary

Model/agent work is grounded in authorized Evidence references, never in ungrounded assertion presented as fact. AI OUTPUT ≠ EVIDENCE (Ch.36 §36.8 final audit) unless an independent governed process establishes an appropriate Evidence object from an actual source/acquisition event. AI transformation of Evidence remains traceable to the original Evidence.

## 18.2 Grounding Package

- Evidence References — the specific authorized Evidence objects consulted.
- Source Authority — provenance of each reference (NCIE-003 Ch.17).
- Contradiction Set — known contradictions among referenced Evidence.
- Citation Metadata — sufficient for a human reviewer to verify the grounding.

Evidence/tool access is governed independently at use time; an AI proposal or verbal request never becomes execution authority (Ch.21).

*Traceability:* NCIE-003 Ch. 17.

# Tool Gateway Integration

Require all consequential tool access through the governed Tool Gateway.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — The Tool Gateway is the single governed entry boundary for consequential tool access; this chapter specifies its contract.

## 19.1 Tool Gateway Mandate

All agent tool use goes through the governed Tool Gateway. No synthesized or registered agent receives direct unrestricted Database, Operating System, Network, Blocking Interface, Tracking Interface or External System access. TOOL ACCESS ≠ ACTION AUTHORITY (front matter §0.3) — possessing a tool binding never itself authorizes a consequential action (Ch.21).

## 19.2 Gateway Contract

Table 13. Tool Gateway request/response contract.

| Element             | Purpose                                                                                 |
|---------------------|-----------------------------------------------------------------------------------------|
| Tool Registry       | Eligible tools and their declared risk class (Ch.20).                                   |
| Authorization Check | Current authorization of the requesting actor (Ch.14) for this specific tool/target.    |
| Target Validation   | Confirms the exact target/action matches what was authorized — never a broader match.   |
| Idempotency         | Correlation key preventing duplicate execution under retry (Ch.30).                     |
| Execution Receipt   | Verifiable confirmation of outcome, or an explicit OUTCOME UNKNOWN state (Ch.30 §30.2). |

Traceability: NCIE-002 Ch.4.

# Read-Only vs State-Changing Tool Classes

Classify tool risk and distinguish sensitive read/reveal from state-changing actions.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-003 Ch.4 — Disclosure-risk classification is canonical; this chapter applies it to tool risk classification.

## 20.1 Read-Only Does Not Mean Harmless

Search, Reveal, Export, Graph Traversal, Protected Identity Retrieval and Precise Location Retrieval are treated as potentially consequential even though they do not mutate state. Tool risk classification reflects disclosure risk, not only mutation risk.

## 20.2 Tool Risk Classes

Table 14. Tool risk classification — disclosure risk and mutation risk are assessed independently.

| Class                           | Examples                                                                                  | Governs                                                                          |
|---------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
| Read-Only, Low Disclosure Risk  | Non-sensitive metadata lookup                                                             | Standard authorization check                                                     |
| Read-Only, High Disclosure Risk | Protected Identity Retrieval, Precise Location, Restricted Graph Traversal, Reveal/Export | Elevated authorization + disclosure-purpose check (NCIE-006 Ch.25 §25.3 pattern) |
| State-Changing, Reversible      | Draft creation, non-consequential updates                                                 | Standard authorization + idempotency (Ch.30)                                     |
| State-Changing, Consequential   | Blocking Interface, Tracking Interface, external regulatory action                        | Current human/institutional approval required before execution (Ch.21)           |

Traceability: NCIE-003 Ch.4.

# Human Approval, Confirmation & Execution Boundary

Define mandatory human/current-authority boundaries before consequential execution.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — Human-Primary governance over consequential execution is architecturally fixed; this chapter specifies the approval sequence.

## 21.1 Consequential-Action Sequence

PROPOSED ≠ AUTHORIZED ≠ APPROVED ≠ EXECUTED ≠ VERIFIED (front matter §0.3). An agent may prepare or recommend an action. Current institutional authority — never Model Eligibility, Agent Registration, VPF Gate A/B PASS, Agent Consensus, Tool Availability, Prior Decision, Institutional Knowledge or Human Presence alone — determines whether execution may proceed (Ch.36 §36.7 final audit).

## 21.2 Execution Boundary Table

Table 15. Consequential-action stage sequence — no stage is skipped for a consequential action.

| Stage      | Meaning                                                                     |
|------------|-----------------------------------------------------------------------------|
| Proposed   | An agent/model has generated a candidate action; no authority attaches.     |
| Authorized | Current governed authorization exists for this specific action/target.      |
| Approved   | The accountable current human/institutional process has approved execution. |
| Executed   | The Tool Gateway has performed the action and returned a receipt.           |
| Verified   | The outcome has been reconciled against the intended result (Ch.30 §30.2).  |

Traceability: NCIE-002 Ch.4.

# Agent-to-Agent Tool Delegation

*Govern indirect agent-to-agent tool delegation without transitive privilege escalation or Agent Factory access.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007-S01 §8 — Recursive synthesis is disabled by default; this chapter applies the same non-escalation discipline to agent-to-agent tool delegation.

## 22.1 Delegation Without Escalation

Synthesized agents do not possess Agent Factory authority by default and do not recursively synthesize agents (Ch.12 §12.3). Where agent-to-agent tool delegation is permitted at all, it remains independently authorized at the final Tool Gateway invocation — an intermediate agent's claimed permission is never trusted in place of that final check.

## 22.2 Delegation Record

If permitted, every agent-to-agent delegation preserves:

- Parent Task; Purpose; Effective Principal.
- Classification; Context Subset.
- Tool Ceiling; Resource Ceiling.
- Provenance.

Delegation cannot create transitive privilege expansion (Ch.14 §14.3).

## Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm whether agent-to-agent tool delegation is permitted in the initial release, and if so its maximum delegation depth.

*Traceability:* NCIE-007-S01 §8.

## Planning, Reflection & Iterative Analytical Loops

*Bound planning/revision loops while persisting defensible artifacts rather than private chain-of-thought.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-006 Ch.11 §11.2 — Decision-Time / private chain-of-thought boundaries are canonical; this chapter bounds planning loops accordingly.

### 23.1 Bounded Planning

Bounded planning/revision loops are permitted where useful, governed by an Iteration Budget, Time Budget, Token Budget, Tool Budget, Termination Conditions and Escalation Conditions (Ch.29 budgets apply identically here).

### 23.2 Persisted vs Private Artifacts

Private chain-of-thought is never persisted. Defensible artifacts persisted instead include:

- Plan State; Questions; Hypotheses.
- Contradictions; Evidence Gaps.
- Tool Results; Final Output.

*Traceability: NCIE-006 Ch.11 §11.2.*

# Prompt, Instruction & Policy Assembly

Define the trusted instruction hierarchy and prompt/policy assembly order.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — Retrieved content ≠ trusted instruction is architecturally fixed; this chapter specifies the assembly hierarchy that enforces it.

## 24.1 Trust Hierarchy

Figure 5. Prompt/Instruction Trust Hierarchy — retrieved content is data; it never gains instruction authority by appearing in a prompt.

|    |                                |
|----|--------------------------------|
| 01 | Governed NCIE Policy           |
|    | ▼                              |
| 02 | Task/Agent Policy              |
|    | ▼                              |
| 03 | Current Authorized User Intent |
|    | ▼                              |
| 04 | Governed Context               |
|    | ▼                              |
| 05 | Retrieved Evidence/Data        |
|    | ▼                              |
| 06 | Tool Output                    |

## 24.2 Assembly Discipline

Prompt/policy assembly follows the hierarchy in §24.1 strictly. A lower-trust element (Retrieved Evidence/Data, Tool Output) is never assembled in a way that lets it override a higher-trust element (Governed NCIE Policy, Task/Agent Policy, Current Authorized User Intent).

Traceability: NCIE-002 Ch.4.

## Prompt Injection & Untrusted-Content Containment

*Contain prompt injection and malicious instructions embedded in retrieved content.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — Retrieved content ≠ trusted instruction is architecturally fixed; this chapter specifies containment testing.

### 25.1 Malicious-Instruction Surfaces

Malicious instructions may originate from Evidence, Documents, Memory, Institutional Knowledge Candidates, Web/API Content, Tool Results or another Agent. Every such surface is explicitly tested; none is assumed safe merely because it passed prior validation for a different purpose. None may override the higher-authority policy levels in Ch.24 §24.1.

### 25.2 Containment Discipline

Content from any of §25.1's surfaces is treated as data throughout assembly (Ch.24), execution (Ch.19-21) and validation (Ch.26). An instruction-shaped string appearing inside retrieved content never gains the authority of a Task/Agent Policy or Current Authorized User Intent element.

*Traceability: NCIE-002 Ch.4.*

# Output Schemas, Validation & the VPF Validation Framework

Define the mandatory VPF Validation Framework — dimensions, profiles and dispositions — that all substantive AI-generated output must enter before validated release, with profile rigor (not exemption) varying by risk.

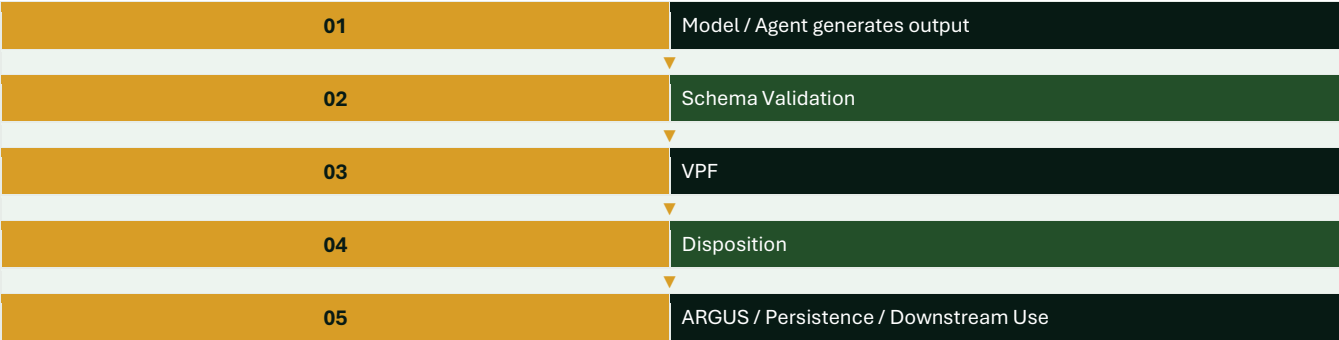
Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007 Skeletal v0.2 — VPF is a mandatory controlled addition approved for inclusion before full expansion; this chapter is its substantive architecture.

**VPF ARCHITECTURE.** ALL SUBSTANTIVE AI-GENERATED OUTPUT SHALL ENTER THE VPF GOVERNANCE PATH BEFORE RELEASE AS A VALIDATED NCIE AI RESULT OR PERSISTENCE AS A GOVERNED AI-GENERATED ANALYTICAL ARTIFACT. The governed VPF profile may vary by output class and risk — including a lightweight deterministic profile for low-risk, mechanical AI output — but no substantive AI output is exempt from VPF by default (§26.5). Purely deterministic, non-AI processing that involves no model/agent inference remains outside this AI-output VPF mandate. No agent type — registered, ephemeral, or ARGUS's own direct model use — receives an implicit exemption.

## 26.1 Normal Path

Figure 6. VPF Normal Path — every substantive AI output traverses schema validation and VPF before validated release.



## 26.2 VPF Dispositions

Table 16. VPF disposition semantics — consistent across every chapter; no chapter redefines them independently.

| Disposition            | Meaning                                                                                                                                                                                                                                                             |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PASS                   | The AI result satisfied the applicable validation framework — not that NCA has institutionally approved the conclusion.                                                                                                                                             |
| QUALIFIED              | The result is releasable with an explicit qualification (Evidence Incomplete, Material Contradiction, Historical Context Only, Current-State Verification Unavailable, Methodological Limitation). Never rendered visually or verbally as equivalent to PASS.       |
| REJECT                 | The result shall not be presented as a validated NCIE AI result. Provenance/audit of the rejected run is retained where policy requires; the system does not silently regenerate until something passes without preserving the failed attempts where consequential. |
| HUMAN_REVIEW_REQUIRED  | ARGUS may present the output only in the explicitly governed review context appropriate to that state — never as validated institutional guidance.                                                                                                                  |
| VALIDATION_UNAVAILABLE | A mandatory validator did not run, timed out, failed technically, or was ineligible. A PASS never occurs on this basis.                                                                                                                                             |

## 26.3 VPF Validation Dimensions

Applicable VPF validation dimensions include, where relevant:

- Evidence grounding; Source/provenance integrity.
- Epistemic classification; Contradiction handling; Assumption exposure.
- Methodological validity; Claim-boundary compliance.
- Temporal validity; Authorization; Classification; Policy compliance.

- Human-Primary boundary; Output-schema integrity.

No single universal numerical VPF score is manufactured across these dimensions — a PASS/QUALIFIED/REJECT/HUMAN\_REVIEW\_REQUIRED disposition is explainable along the specific dimensions that produced it.

## 26.4 Two Gates

VPF GATE A validates the dynamically synthesized Agent Envelope before execution (Ch.11 §11.5). VPF GATE B validates substantive AI output after execution and before validated release. Neither gate substitutes for current human/institutional authorization where required (Ch.21). The two gates are never collapsed.

## 26.5 Validation Profiles

Different AI tasks require different validation profiles — for example Question Answer, Hypothesis, Contradiction, Evidence-Gap Analysis, Cross-Domain Analysis, Briefing, Recommendation, Tool Proposal and Dynamic-Agent Output. Every profile has a defined Purpose, Applicable Scope, Version and Lifecycle State, identifies its mandatory validation dimensions, and cannot silently weaken global mandatory controls — a weaker profile never bypasses mandatory authority/security controls.

SUBSTANTIVE AI OUTPUT ≠ VPF-EXEMPT BY DEFAULT (front matter §0.3). A low-risk, mechanical AI output (for example a routine reformatting or extraction task with negligible epistemic or authority consequence) may be governed by a lightweight, largely deterministic VPF profile with correspondingly narrow mandatory dimensions — it is never routed around VPF altogether merely for latency or usability. Purely deterministic, non-AI processing that involves no model/agent inference is outside the AI-output VPF mandate entirely, since it is not AI-generated output (§26 vpf callout).

## 26.6 Validator Kinds

The architecture distinguishes Deterministic Validators from AI-Assisted Validators. AI is not used for controls that are authoritatively determinable through structured policy/state where deterministic validation is sufficient.

## 26.7 VPF Does Not Become Institutional Authority

VPF validates AI output. VPF does not approve regulatory Decisions, create Findings, authorize blocking, authorize tracking, approve Institutional Knowledge, or replace human judgment. VPF VALIDATION ≠ HUMAN / INSTITUTIONAL APPROVAL — this is a foundational NCIE-007 principle restated, never contradicted, in any chapter.

## 26.8 Human Modification After Validation

If a human materially modifies an AI result after VPF validation, the modified artifact does not automatically inherit the original VPF status. Policy determines whether the modification requires Revalidation or a Human-Origin/Human-Modified designation; both AI and human provenance are preserved regardless.

## 26.9 VPF Provenance

Persist for every consequential VPF validation:

- VPF Validation ID; Validation Profile; Input Artifact Reference.
- Applicable Framework/Policy Version; Validation Time.
- Disposition; Qualification/Failure Reasons.
- Human Review Outcome, where applicable.

## Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Approve the VPF validation dispositions, mandatory dimensions and release behavior for PASS, QUALIFIED, REJECT and HUMAN\_REVIEW\_REQUIRED.
- **[PROPOSED DESIGN DEFAULT]** The disposition set in §26.2 (PASS/QUALIFIED/REJECT/HUMAN\_REVIEW\_REQUIRED/VALIDATION\_UNAVAILABLE) is proposed as complete for launch.

*Traceability: NCIE-007 Skeletal v0.2 — MANDATORY VPF ADDITION, Chapter 26.*

# AI Provenance, Attribution & Execution Receipts

*Preserve end-to-end model, agent, tool, VPF and human attribution, including for Dynamic Agent Synthesis runs.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-005 Ch.27 — Human/ARGUS/workflow attribution is canonical; this chapter extends it to models, agents and VPF.

## 27.1 End-to-End Attribution

Every consequential output/action preserves attribution across Run ID, Actor (Ch.3), Model Route, Agent, Context References, Tool Calls, Output, Human Response and Timestamps. Audit identifies exactly which actor performed each operation (Ch.3 §3.1).

## 27.2 Dynamic-Agent Provenance

For every synthesized agent run, preserve at least:

- Agent Definition ID/version; Synthesis Origin; ARGUS Request.
- VPF Gate A Result; Sandbox/Evaluation Result.
- Runtime Identity; Model Route; Context References; Tool Calls.
- VPF Gate B Result; Output.
- Expiry State; Promotion State.

## 27.3 Artifact Integrity

Agent Definition Packages support integrity controls such as Hash, Signature, Version and Immutable Run Reference where appropriate. An agent definition does not change silently between validation and execution.

*Traceability: NCIE-005 Ch.27.*

# Safety Guardrails & Policy Enforcement

*Enforce technical safety policy beyond prompt instructions.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — AI safety and service-boundary requirements are architecturally fixed; this chapter specifies technical enforcement beyond prompt instructions.

## 28.1 Enforcement Beyond Prompt Instructions

Safety policy is enforced technically — through the Model Gateway, Tool Gateway, VPF and Agent Factory controls — never relying solely on prompt-level instruction to models or agents, which retrieved content can attempt to override (Ch.25).

*Traceability: NCIE-002 Ch.4.*

## Resource Budgets, Cost, Rate Limits & Quotas

*Bound token, time, concurrency, cost and tool-call consumption for every model/agent invocation.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-004 Ch.31 — Performance/tiering targets are engineering-governed; this chapter specifies orchestration-level resource bounding.

### 29.1 Bounded Resources

Every model/agent invocation, including every synthesized agent, requires bounded:

- Runtime; Tokens; Model Calls; Tool Calls.
- Concurrency; Memory/Context Volume.
- Compute/GPU, where relevant.

Runaway agent loops terminate safely rather than running unbounded.

### 29.2 Cancellation

Humans and governed orchestration can cancel active agent runs where appropriate. Cancellation does not leave consequential tool operations in an ambiguous state; reconciliation (Ch.30 §30.2) applies when an external outcome is unknown.

*Traceability:* NCIE-004 Ch.31.

# Concurrency, Idempotency & Duplicate Suppression

*Prevent duplicate actions and stale-result overwrites under retries/concurrency.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-002 Ch.4 — Zero-trust concurrency handling is architecturally fixed; this chapter specifies idempotency and unknown-outcome handling.

## 30.1 Duplicate Suppression

Idempotency and correlation apply to Agent Runs, Tool Calls, Action Requests and VPF Validation Requests wherever retries are possible.

## 30.2 Unknown Execution Outcome

UNKNOWN OUTCOME ≠ FAILURE and UNKNOWN OUTCOME ≠ SUCCESS (front matter §0.3). A consequential operation with an unknown external outcome is never blindly retried; it triggers reconciliation instead. TIMEOUT ≠ FAILURE and TIMEOUT ≠ SUCCESS follow the same rule.

## 30.3 Stale Agent Results

If authoritative context changes materially while an agent executes, its result may require STALE\_CONTEXT, REVALIDATION\_REQUIRED or HUMAN\_REVIEW\_REQUIRED. Newer authoritative state is never overwritten by stale agent output.

*Traceability:* NCIE-002 Ch.4.

# Observability, Telemetry & AI Operations

*Provide operational visibility into routing, agents, tools, VPF, Agent Factory and failures.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-006 Ch.31 §31.3 — Technical-vs-governance observability separation is canonical; this chapter applies it to orchestration.

## 31.1 Separate Operational Visibility

Separate operational visibility exists for the Model Gateway, Agent Factory, Agent Runtime, Tool Gateway, VPF, the Registries (Ch.6, Ch.12 §12.2) and ARGUS. Route/agent/tool metrics, latency, token/cost, policy denials, error taxonomy and distributed trace are captured per component, not collapsed into one undifferentiated stream.

## 31.2 Telemetry Privacy

Telemetry does not unnecessarily expose Prompts, Evidence, Protected Identity, Secrets, Precise Location or Sensitive Tool Results.

## 31.3 Telemetry Is Not Audit

TELEMETRY ≠ AUDIT. Operational telemetry (§31.1) is never mistaken for the authoritative provenance/audit architecture (Ch.27); the two serve different purposes and different retention rules.

*Traceability: NCIE-006 Ch.31 §31.3.*

# Evaluation, Regression & Adversarial Testing

Define release evaluation, regression and adversarial testing for orchestration, VPF and Dynamic Agent Synthesis.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-003 Ch.38 — Dedicated temporal/lineage/authorization test categories are canonical; this chapter is their orchestration, VPF and Dynamic Agent application.

## 32.1 Evaluation Programme Coverage

The evaluation programme includes at minimum:

- Positive Capability Tests; Negative Tests; Adversarial Tests; Regression.
- Model Replacement; Provider Replacement.
- Agent Factory; VPF; Dynamic Agents.
- Human-Primary Authority.

## 32.2 Mandatory Acceptance Test Matrix

Every named test below is mandatory before NCIE-007 v1.0 delivery. Categories are condensed here per the token-conservation convention (front matter §0.2); the full test set traces one-to-one to the handover instruction's acceptance-test enumeration.

Table 17. Mandatory Acceptance Test Matrix — condensed per category; every listed capability/boundary must fail safely under its adversarial test.

| Category                          | Representative Tests                                                                                                                       | Must Demonstrate                                                                                                                                                                                                                                        |
|-----------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VPF temporal/current-state        | Decision-Time contamination test; current-state reconciliation test                                                                        | Later Evidence/Knowledge/Rules never contaminate historical reconstruction; historical Memory used for a current claim without reconciliation yields QUALIFIED/HUMAN_REVIEW_REQUIRED, never PASS                                                        |
| VPF authorization/disclosure      | Restricted-information leakage test; spoken-disclosure test                                                                                | VPF never reveals restricted information, never treats validation as disclosure authorization; VPF PASS ≠ spoken-disclosure authorization                                                                                                               |
| VPF non-authority                 | Human-approval test; Knowledge/Finding/Decision promotion tests                                                                            | A VPF-PASS recommendation for a consequential action remains PROPOSED/AWAITING AUTHORIZATION; VPF-PASS artifacts cannot auto-promote to Institutional Knowledge, Finding or Decision                                                                    |
| Dynamic-agent scope/authorization | Envelope-expansion test; cross-user reuse test; cross-context retrieval test                                                               | Every unauthorized expansion beyond the validated envelope is denied; historical/cached authority is never inherited across users or contexts                                                                                                           |
| Dynamic-agent governance boundary | Self-registration test; recursive-synthesis test; Gate B/Tool Gateway/Model Gateway/Registry bypass tests; VPF tampering test              | Self-registration denied; recursive synthesis denied by default; no bypass path exists for any gate or registry; VPF cannot be disabled, weakened or forged                                                                                             |
| Dynamic-agent resource/lifecycle  | Resource-ceiling test; expiry test; reactivation test; promotion/promotion-rejection tests; duplicate-capability test; capability-gap test | Enforced termination at budget limits; expired agents do not silently reactivate with stale authorization; only governed human/institutional authority approves registration; missing capability yields ENGINEERING CHANGE CANDIDATE, never fabrication |
| Generated-code boundary           | Positive sandboxed-execution test; negative escape/exfiltration test                                                                       | If unapproved for initial release, no path silently activates it; if approved, sandbox/network/secret/dependency controls hold under adversarial attempt                                                                                                |

| Category                       | Representative Tests                                                                                                                                                                     | Must Demonstrate                                                                                                                                                  |
|--------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Compromise/suspension response | Agent Factory, VPF validator, model and tool compromise tests; authorization-revocation test; Room-membership-change test                                                                | New activity stops; dependents are identified; historical outputs remain traceable rather than deleted; revoked authorization is not honored through cached state |
| Change management              | Evidence-correction, Knowledge-retraction, Rule-change, model-change, VPF-profile-change, Agent-Definition-change, Agent-Pattern-change, Tool-schema-change, Context-schema-change tests | Dependent artifacts are identified and revalidated/flagged; historical artifacts retain their original reference; nothing is silently presented as current        |
| Fail-closed                    | VPF/Gate A/Model Registry/Tool Registry/human-approval/audit/provenance unavailability tests                                                                                             | A PASS or an executed consequential action never occurs solely because a mandatory control did not run                                                            |
| Sovereignty                    | Dynamic Agent, VPF, Institutional Memory, Agent Registry, Tool, human-authority sovereignty tests                                                                                        | Each capability/doctrine survives replacement of the underlying model, provider, runtime or framework                                                             |
| Human-Primary final scenario   | Full-chain adversarial scenario (Gate A/B pass, sandbox pass, agents agree, Knowledge supports the recommendation, tool available)                                                       | None of those facts alone creates institutional execution authority absent required current human/institutional authorization — mandatory before delivery         |

32.3 Adversarial Envelope Tests

Agent-envelope tests specifically probe authority expansion, unsafe tool/data access, recursive synthesis and excessive resource scope. Output tests specifically verify that unvalidated or rejected results cannot be released as validated NCIE AI output.

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm required test environments and masked/synthetic test data provisioning for the acceptance test matrix, consistent with NCIE-003 Ch.29 §29.3's convention.

Traceability: NCIE-003 Ch.38.

# Model & Agent Lifecycle Governance

Govern model/agent onboarding, approval scope, suspension, deprecation, retirement and promotion.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007-S01 §5 — The DRAFT→ACTIVE\_REUSABLE lifecycle is the normative supplement baseline; this chapter integrates it with Model lifecycle governance.

## 33.1 Unified Lifecycle Distinctions

State machines remain distinct across:

- Model; Agent Definition; Ephemeral Agent; Registered Agent.
- Agent Run; Task; Promotion Candidate.
- Tool Invocation; VPF Profile; VPF Validation; Human Approval.

Shared storage implementation never collapses these distinct governance semantics (Ch.12 §12.2).

## 33.2 Model Lifecycle

Table 18. Model lifecycle states.

| State                | Meaning                                                                  |
|----------------------|--------------------------------------------------------------------------|
| Candidate            | Proposed for evaluation; not yet eligible for routing.                   |
| Evaluated            | Evaluation complete; approval pending.                                   |
| Approved / Active    | Eligible for routing per Ch.6-7.                                         |
| Restricted           | Eligible only for a narrowed scope.                                      |
| Suspended            | Temporarily ineligible even if technically available (Ch.10 §10.2).      |
| Deprecated / Retired | No longer eligible for new routing; historical outputs remain traceable. |

## 33.3 Change-Management Discipline

A model, VPF profile, Agent Definition or Agent Pattern change produces a new version; existing historical instances retain their original version/profile/pattern reference while new instances use the current eligible version. A material Agent Definition modification triggers new-version creation plus applicable Evaluation, Gate A revalidation and Approval.

### Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm model-approval authority and emergency model/agent suspension authority.

Traceability: NCIE-007-S01 §5.

## Security, Isolation & Supply-Chain Boundaries

*Define orchestration-specific isolation and supply-chain boundaries without duplicating NCIE-009.*

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007-S01 §6 — Runtime and sandbox boundary requirements are the normative supplement baseline; this chapter details sandbox evaluation without duplicating NCIE-009.

### 34.1 Runtime & Sandbox Boundary

Synthesized agents execute in bounded runtime isolation with explicit tool, data, model, network, context and resource ceilings. They cannot grant themselves new capabilities or bypass the Tool Gateway, Model Gateway, Memory/Context services, VPF or current authorization.

### 34.2 Sandbox Evaluation Requirements

Before ephemeral activation, evaluate the synthesized agent in isolation appropriate to its risk, testing:

- Schema Compliance; Tool Restrictions; Data-Class Restrictions.
- Network/Egress; Resource Limits; Termination.
- Prompt Injection; Authority Expansion; VPF Integration.

Sandbox success does not imply Registration, Institutional Approval, Output Validity or Action Authorization.  
SANDBOX PASS ≠ ACTIVATION / REGISTRATION / OUTPUT APPROVAL.

### 34.3 Deterministic Continuity

AI orchestration failure does not unnecessarily disable deterministic NCIE functions — a core resilience requirement (Ch.10 §10.1).

### 34.4 Generated-Code Boundary

If generated executable code remains outside the approved initial-release scope, this is marked GENERATED-CODE AGENT ACTIVATION — NOT YET APPROVED / SECURITY & ENGINEERING CONFIRMATION REQUIRED. The broader Dynamic Agent Synthesis approval never silently activates generated-code capability. If later approved, the specification requires Static Analysis, Dependency Control, Sandbox, Network Restrictions, Secrets Protection, Runtime Limits, Artifact Integrity, Evaluation and an Engineering Promotion Boundary.

### Human Review Focus

- **[SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED]** Determine the initial disposition of generated executable code: out of initial scope, proposed with restrictions, or approved for a defined risk class. Blocking only for generated-code agent activation, not for configuration-only Dynamic Agent Synthesis.

*Traceability:* NCIE-007-S01 §6.

# Human Review Register, Open Decisions & Proposed Defaults

Consolidate every unresolved orchestration, VPF and Dynamic Agent Synthesis decision with category and blocking state.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007 — — Consolidates every Human Review Focus item from Chapters 1-34 into one authoritative register.

## 35.1 Purpose

Consolidates every Human Review Focus item raised in Chapters 1-34, extracted directly from the chapter review blocks so this register cannot drift from its sources. Each item states precisely what it blocks; no unresolved optional capability is allowed to imply that the whole NCIE-007 architecture is unusable (handover items 1312/1370/1385).

## 35.2 Summary

Table 19. Human Review Register summary counts.

| Metric                                        | Count |
|-----------------------------------------------|-------|
| Total items                                   | 15    |
| Resolved by Upstream Baseline                 | 1     |
| Proposed Design Default                       | 2     |
| Institutional Confirmation Required           | 8     |
| Security/Sovereignty Confirmation Required    | 3     |
| Legal/Policy Confirmation Required            | 0     |
| Engineering/Infrastructure Discovery Required | 1     |
| Source/Workflow Discovery Required            | 0     |
| Blocking                                      | 12    |
| Non-Blocking                                  | 3     |

## 35.3 Complete Register

Table 20. Complete Human Review Register (Chapters 1-34).

| ID       | Ch.    | Issue                                                                                                                                                                                                          | Category                                   | Blocking?    |
|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------|--------------|
| HR7-1-1  | Ch.1   | Confirm the acceptance authority for NCIE-007 as a whole and for individual chapter sign-off.                                                                                                                  | INSTITUTIONAL CONFIRMATION REQUIRED        | BLOCKING     |
| HR7-2-1  | Ch.2   | All 19 governing invariants are fixed by NCIE-002 Ch.4 and the approved NCIE-007 Skeletal v0.2; no institutional decision is required to adopt them.                                                           | RESOLVED BY UPSTREAM BASELINE              | NON-BLOCKING |
| HR7-6-1  | Ch.6   | Confirm the model-approval authority and cadence for Registry review.                                                                                                                                          | INSTITUTIONAL CONFIRMATION REQUIRED        | BLOCKING     |
| HR7-11-1 | Ch.1 1 | Approve the ephemeral-agent activation policy by task/risk class — whether ephemeral agents may activate automatically after Gate A plus security/evaluation checks, or require human approval per risk class. | INSTITUTIONAL CONFIRMATION REQUIRED        | BLOCKING     |
| HR7-11-2 | Ch.1 1 | Approve the promotion authority and criteria for converting an ephemeral agent into a registered reusable institutional agent.                                                                                 | INSTITUTIONAL CONFIRMATION REQUIRED        | BLOCKING     |
| HR7-12-1 | Ch.1 2 | Approve the Agent Factory resource, model, tool, data and network ceilings applied to every synthesized agent.                                                                                                 | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED | BLOCKING     |

| ID       | Ch.    | Issue                                                                                                                                                                                                                                                          | Category                                      | Blocking?    |
|----------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|--------------|
| HR7-12-2 | Ch.1 2 | Recursive Agent Synthesis remains disabled by default pending any future explicit governed policy change, consistent with NCIE-007-S01 §8.                                                                                                                     | PROPOSED DESIGN DEFAULT                       | NON-BLOCKING |
| HR7-14-1 | Ch.1 4 | Confirm the authorization-revocation propagation latency acceptable for an active agent run — historical authorization shall not remain effective through cached agent state.                                                                                  | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED    | BLOCKING     |
| HR7-16-1 | Ch.1 6 | Confirm checkpoint retention duration and storage tier separate from institutional Memory retention (NCIE-006 Ch.28).                                                                                                                                          | ENGINEERING/INFRASTRUCTURE DISCOVERY REQUIRED | BLOCKING     |
| HR7-22-1 | Ch.2 2 | Confirm whether agent-to-agent tool delegation is permitted in the initial release, and if so its maximum delegation depth.                                                                                                                                    | INSTITUTIONAL CONFIRMATION REQUIRED           | BLOCKING     |
| HR7-26-1 | Ch.2 6 | Approve the VPF validation dispositions, mandatory dimensions and release behavior for PASS, QUALIFIED, REJECT and HUMAN_REVIEW_REQUIRED.                                                                                                                      | INSTITUTIONAL CONFIRMATION REQUIRED           | BLOCKING     |
| HR7-26-2 | Ch.2 6 | The disposition set in §26.2 (PASS/QUALIFIED/REJECT/HUMAN_REVIEW_REQUIRED/VALIDATION_UNAVAILABLE) is proposed as complete for launch.                                                                                                                          | PROPOSED DESIGN DEFAULT                       | NON-BLOCKING |
| HR7-32-1 | Ch.3 2 | Confirm required test environments and masked/synthetic test data provisioning for the acceptance test matrix, consistent with NCIE-003 Ch.29 §29.3's convention.                                                                                              | INSTITUTIONAL CONFIRMATION REQUIRED           | BLOCKING     |
| HR7-33-1 | Ch.3 3 | Confirm model-approval authority and emergency model/agent suspension authority.                                                                                                                                                                               | INSTITUTIONAL CONFIRMATION REQUIRED           | BLOCKING     |
| HR7-34-1 | Ch.3 4 | Determine the initial disposition of generated executable code: out of initial scope, proposed with restrictions, or approved for a defined risk class. Blocking only for generated-code agent activation, not for configuration-only Dynamic Agent Synthesis. | SECURITY/SOVEREIGNTY CONFIRMATION REQUIRED    | BLOCKING     |

### 35.4 Mechanical Verification

Per handover items 1311/1369, this register's totals are verified programmatically, not by inspection: the production build independently re-extracts every ('review', ...) block across Chapters 1-34 and asserts the result matches this chapter's own extraction (§35.2-§35.3) exactly, by category and by blocking status; a mismatch fails the build rather than reaching this document. The current build confirms an exact match — 15 items (1 Resolved / 2 Proposed / 8 Institutional / 3 Security / 0 Legal / 1 Infrastructure / 0 Workflow), 12 Blocking / 3 Non-Blocking. This figure and §35.2's summary table are generated from the same computed extraction at build time, so they cannot diverge from one another.

### 35.5 Blocking Scope Discipline

Where a Blocking item remains open, it blocks only the specific capability it names — for example, generated-code Dynamic Agent activation (Ch.34 §34.4) remains blocked pending Security/Engineering confirmation while configuration-only Dynamic Agent Synthesis (Ch.11) remains fully specified and reviewable. No Blocking item is read as blocking the entire NCIE-007 architecture.

*Traceability: Consolidates Chapters 1-34; individual chapters trace to NCIE-002/003/004/005/006/007-S01 as cited therein.*

# Requirements Traceability, Acceptance & Handover Gate

Provide the complete upstream crosswalk, Upstream Impact Register, Downstream Handover Register, acceptance scenarios and controlled handover to NCIE-008.

Status: IN DEVELOPMENT — FOR HUMAN REVIEW.

**Upstream:** NCIE-007 — — Closes the authority chain: NCIE-001 requirements → ... → NCIE-006 Memory & Context → NCIE-007 AI/Agent/Model Orchestration.

## 36.1 Traceability Matrix

Table 21. NCIE-007 chapter-to-upstream traceability matrix.

| Ch. | Title                                                    | Primary Upstream Dependency               |
|-----|----------------------------------------------------------|-------------------------------------------|
| 1   | Mandate, Authority Chain & Specification Boundary        | NCIE-002 Ch.4, Ch.24; NCIE-005; NCIE-006  |
| 2   | Orchestration Doctrine & Governing Invariants            | NCIE-002 Ch.4; NCIE-006 front matter §0.3 |
| 3   | AI Runtime Actor & Capability Taxonomy                   | NCIE-002 Ch.4                             |
| 4   | ARGUS Orchestrator Boundary                              | NCIE-005 Ch.4                             |
| 5   | Model Gateway Contract                                   | NCIE-002 Ch.24                            |
| 6   | Approved Model Registry                                  | NCIE-004 Ch.18                            |
| 7   | Classification-, Purpose- & Task-Aware Model Routing     | NCIE-003 Ch.31                            |
| 8   | Provider Adapters & Provider Neutrality                  | NCIE-004 Ch.18                            |
| 9   | Model Selection, Versioning & Reproducibility            | NCIE-003 Ch.28                            |
| 10  | Fallback, Failover & Degraded AI Operation               | NCIE-002 Ch.24                            |
| 11  | Specialized Agent Architecture & Dynamic Agent Synthesis | NCIE-007-S01 §3-5; NCIE-005 Ch.4          |
| 12  | Agent Capability Registry & Agent Factory                | NCIE-007-S01 §1, §7-8                     |
| 13  | Task Decomposition & Delegation                          | NCIE-002 Ch.4                             |
| 14  | Delegated Authorization & Identity Propagation           | NCIE-006 Ch.25 §25.3                      |
| 15  | Multi-Agent Coordination & Conflict Handling             | NCIE-002 Ch.4                             |
| 16  | Agent State, Checkpointing & Resumption                  | NCIE-006 Ch.20                            |
| 17  | Context Acquisition & Memory-Service Integration         | NCIE-006 Ch.14, Ch.17                     |
| 18  | Evidence & Grounding Integration                         | NCIE-003 Ch.17                            |
| 19  | Tool Gateway Integration                                 | NCIE-002 Ch.4                             |
| 20  | Read-Only vs State-Changing Tool Classes                 | NCIE-003 Ch.4                             |
| 21  | Human Approval, Confirmation & Execution Boundary        | NCIE-002 Ch.4                             |
| 22  | Agent-to-Agent Tool Delegation                           | NCIE-007-S01 §8                           |
| 23  | Planning, Reflection & Iterative Analytical Loops        | NCIE-006 Ch.11 §11.2                      |
| 24  | Prompt, Instruction & Policy Assembly                    | NCIE-002 Ch.4                             |
| 25  | Prompt Injection & Untrusted-Content Containment         | NCIE-002 Ch.4                             |

| Ch. | Title                                                     | Primary Upstream Dependency                           |
|-----|-----------------------------------------------------------|-------------------------------------------------------|
| 26  | Output Schemas, Validation & the VPF Validation Framework | NCIE-007 Skeletal v0.2 — MANDATORY VPF ADDITION       |
| 27  | AI Provenance, Attribution & Execution Receipts           | NCIE-005 Ch.27                                        |
| 28  | Safety Guardrails & Policy Enforcement                    | NCIE-002 Ch.4                                         |
| 29  | Resource Budgets, Cost, Rate Limits & Quotas              | NCIE-004 Ch.31                                        |
| 30  | Concurrency, Idempotency & Duplicate Suppression          | NCIE-002 Ch.4                                         |
| 31  | Observability, Telemetry & AI Operations                  | NCIE-006 Ch.31 §31.3                                  |
| 32  | Evaluation, Regression & Adversarial Testing              | NCIE-003 Ch.38                                        |
| 33  | Model & Agent Lifecycle Governance                        | NCIE-007-S01 §5                                       |
| 34  | Security, Isolation & Supply-Chain Boundaries             | NCIE-007-S01 §6                                       |
| 35  | Human Review Register & Open Decisions                    | Consolidates Ch.1-34                                  |
| 36  | Traceability, Acceptance & Handover                       | Closes NCIE-001→...→NCIE-006→NCIE-007 authority chain |

36.2 Traceability Mechanical Verification

Per handover items 1307/1371, every one of the 36 chapters above carries at least one upstream citation, verified programmatically at build time (not merely by inspection of §36.1) against every chapter's own ('upstream', ...) block; a build with any chapter missing one fails before reaching this document. No ORPHAN NCIE-007 REQUIREMENT was found in this edition.

36.3 Upstream Impact Register

NCIE-001–006 remain approved and are not amended during this production cycle (front matter §0.1). The following impacts are recorded for the surgical amendment pass that NCIE-007 approval shall initiate:

Table 22. VPF & Dynamic Agent Upstream Impact Register — controlled for post-NCIE-007 surgical incorporation; this is a documentation-governance action, not an amendment performed now.

| Impact ID | Affected Document | Required Surgical Amendment                                                                                                                         | Severity | Blocking Scope                   | Amendment Status                 |
|-----------|-------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------|----------------------------------|
| UIR-001   | NCIE-001          | Recognize bounded Dynamic Agent Synthesis and VPF-validated AI output as explicit NCIE capabilities.                                                | Material | Does not block NCIE-007 approval | DEFERRED UNTIL NCIE-007 APPROVAL |
| UIR-002   | NCIE-002          | Add Dynamic Agent Factory/synthesis boundary, ephemeral-vs-registered lifecycle, and VPF validation gates without changing Human-Primary authority. | Material | Does not block NCIE-007 approval | DEFERRED UNTIL NCIE-007 APPROVAL |
| UIR-003   | NCIE-003          | Add/extend Agent Definition, Agent Instance, VPF Validation Result, Promotion Candidate and agent lifecycle/provenance data semantics.              | Material | Does not block NCIE-007 approval | DEFERRED UNTIL NCIE-007 APPROVAL |

| Impact ID | Affected Document | Required Surgical Amendment                                                                                                                                              | Severity | Blocking Scope                   | Amendment Status                 |
|-----------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------------------------------|----------------------------------|
| UIR-004   | NCIE-004          | Add sandboxed Agent Factory/runtime, agent-definition artifact controls, evaluation pipeline, signing/integrity and VPF integration as governed technology capabilities. | Material | Does not block NCIE-007 approval | DEFERRED UNTIL NCIE-007 APPROVAL |
| UIR-005   | NCIE-005          | Allow ARGUS to initiate bounded agent synthesis when needed; confirm ARGUS still cannot grant authority, self-register an agent, or bypass human approval.               | Material | Does not block NCIE-007 approval | DEFERRED UNTIL NCIE-007 APPROVAL |
| UIR-006   | NCIE-006          | Confirm synthesized agents consume governed minimum-necessary Context/Memory and receive no unrestricted Memory or inherited ARGUS-wide context.                         | Material | Does not block NCIE-007 approval | DEFERRED UNTIL NCIE-007 APPROVAL |

36.4 Downstream Handover Register

Table 23. Downstream Handover Register — NCIE-007 specifies logical requirements; it does not pre-design these downstream implementations (front matter §0.2).

| Downstream Document                       | NCIE-007 Obligation to Implement                                                                                                                                          |
|-------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| NCIE-008 (Governance & Human Authority)   | Institutional acceptance authority for NCIE-007 chapters and for VPF/Dynamic Agent governance decisions (Ch.1 §1.3, Ch.35).                                               |
| NCIE-009 (Security/IAM)                   | Sandbox isolation, current-authorization enforcement, and Agent Factory/Model/Tool Gateway security implementation (Ch.34 references, never duplicates, this obligation). |
| NCIE-010 (Evidence, Provenance & Audit)   | Physical audit/provenance store implementing the logical fields specified in Ch.27, Ch.26 §26.9.                                                                          |
| NCIE-011 (API/Integration)                | Wire-protocol implementation of the Model Gateway, Tool Gateway and VPF service contracts specified logically in Ch.5, Ch.19, Ch.26.                                      |
| NCIE-012 (UX)                             | User-facing presentation of VPF dispositions, agent status and human approval checkpoints specified logically in Ch.21, Ch.26 §26.2.                                      |
| NCIE-014 (Database & Storage Engineering) | Physical storage for the Model/Agent/Tool/VPF Profile/Agent Pattern Registries and VPF artifact persistence specified logically in Ch.6, Ch.12 §12.2, Ch.26 §26.9.        |
| NCIE-015 (Deployment/Infrastructure)      | Physical sandbox/runtime infrastructure and network/egress enforcement specified logically in Ch.34.                                                                      |
| NCIE-016 (Testing)                        | Full implementation of the Mandatory Acceptance Test Matrix specified in Ch.32 §32.2.                                                                                     |

36.5 Acceptance Checklist

- All 19 Governing Invariants (front matter §0.3) are preserved without exception across all 36 chapters.
- VPF is fully integrated: every substantive AI-generated output enters the VPF governance path (Gate B) before validated release, with profile rigor — never exemption — varying by risk/output class; no chapter implies Model Output can be directly validated by ARGUS, or that registered/ephemeral agents are exempt from VPF by default.

- Dynamic Agent Synthesis is fully integrated: ARGUS can create bounded task-specific agents from approved primitives without Codex hand-coding each one, while agent synthesis cannot create new institutional authority, infrastructure or unapproved capability.
- NCIE-007-S01 is fully crosswalked: every normative S01 requirement is incorporated, retained or explicitly superseded with rationale (§36.6).
- The Human Review Register (Ch.35) is mechanically reconciled (§35.4); the Upstream Impact Register (§36.3) and Downstream Handover Register (§36.4) are complete.
- Required diagrams, matrices, state models and logical contracts are present (Dynamic Agent Synthesis Flow Ch.11 §11.2, Ephemeral Agent Lifecycle Ch.11 §11.7, ARGUS Orchestration Request Lifecycle Ch.4 §4.2, VPF Normal Path Ch.26 §26.1, AI Runtime Actor Taxonomy Ch.3 §3.1, Prompt/Instruction Trust Hierarchy Ch.24 §24.1).
- Acceptance-test coverage (Ch.32 §32.2) addresses VPF, Dynamic Agent Synthesis, sovereignty, fail-closed, change-management and the mandatory Human-Primary final scenario.
- No chapter fully specifies NCIE-008/009/010/011/012/014/015/016; each cross-references the downstream owner (§36.4).
- No technology named from NCIE-004 is asserted as more approved here than its NCIE-004 Technology Registry status.

36.6 S01 Disposition

Every normative NCIE-007-S01 requirement is incorporated into this document (Ch.11-12, Ch.22, Ch.27 §27.2, Ch.33-34) with no silent loss. Proposed disposition: NCIE-007-S01 — FULLY INCORPORATED; CANDIDATE FOR SUPERSESSSION UPON HUMAN APPROVAL OF NCIE-007. Final supersession is not decided here; it requires human approval.

36.7 Readiness Statements

Table 24. Readiness statements — architecture completeness is distinct from operational activation.

| Capability                       | Status                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VPF                              | ARCHITECTURALLY SPECIFIED — IMPLEMENTATION-READY SUBJECT TO OPEN DECISIONS (Ch.35). Not yet operational merely because the specification exists.                                                                                                                                                                                                                |
| Dynamic Agent Synthesis          | ARCHITECTURE SPECIFIED. Agent Factory implementation and Dynamic Agent production activation are separate, later milestones.                                                                                                                                                                                                                                    |
| Codex/engineering pre-conditions | Agent Factory, Agent Definition Schema, Runtime Policy Enforcement, Sandbox/Evaluation, Agent Registry Integration, Model Gateway Integration, Tool Gateway Integration, VPF Gate A/B, Provenance, Observability, Expiry and Kill/Suspend Controls must be implemented and operationally approved before ARGUS exercises Dynamic Agent Synthesis operationally. |

36.8 Capability and Limitation Statement

Once the Ch.36 §36.7 pre-conditions are implemented and approved: ARGUS MAY SYNTHESIZE AND USE BOUNDED TASK-SPECIFIC AI AGENTS FROM APPROVED NCIE PRIMITIVES WITHOUT REQUIRING CODEX TO HAND-CODE EACH INDIVIDUAL AGENT. Simultaneously: ARGUS MAY NOT USE DYNAMIC AGENT SYNTHESIS TO CREATE NEW INSTITUTIONAL AUTHORITY, BYPASS GOVERNED CONTROLS, SELF-REGISTER AGENTS, MODIFY VPF, OR CREATE INFRASTRUCTURE CAPABILITIES THAT DO NOT EXIST WITHIN THE APPROVED SYNTHESIS ENVELOPE. CODEX / ENGINEERING RE-ENTERS WHEN THE REQUIRED CAPABILITY EXCEEDS THE APPROVED AGENT FACTORY PRIMITIVES OR REQUIRES MATERIAL PRODUCTION ENGINEERING CHANGE. Both the capability and the limitation are true simultaneously (Ch.12 §12.1).

36.9 Final Disposition

NCIE-007 v1.0 is architecturally complete across all 36 chapters. This is distinct from all implementation decisions being resolved: Chapter 35 records genuine open decisions, none of which is presented as already resolved. Claude does not recommend approval merely because the document is complete; this document presents sufficient evidence for human review, and formal approval remains with the authorized human governance process (NCIE-008).

Human Review Focus

- **[INSTITUTIONAL CONFIRMATION REQUIRED]** Confirm the boundary with NCIE-008 (Governance & Human Authority) and NCIE-009 (Security/IAM) for the specific acceptance items in §36.5.

*Traceability: Closes the NCIE-001 → NCIE-002 → NCIE-003 → NCIE-004 → NCIE-005 → NCIE-006 → NCIE-007 authority chain (front matter §0.1).*